

MediBook

WhatsApp Dental Appointment Booking Platform

Developed & Powered by **Swalambha AI Technologies**

A complete SaaS solution that lets patients book, reschedule, and cancel dental appointments through WhatsApp — no app download required.



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1. Executive Overview

MediBook eliminates phone-queue frustration for patients and reduces front-desk workload for clinics — all through the world's most popular messaging app.

MediBook is a **cloud-based, multi-tenant SaaS platform** purpose-built for Indian dental clinics. It connects patients to dental providers through **WhatsApp** — a channel already used by over 500 million people in India — without requiring any app download, account creation, or phone calls.

Clinic administrators manage everything — doctors, schedules, appointments, patients, and analytics — through a clean, browser-based dashboard. The platform runs independently for each clinic, so data is completely isolated and private.

14

Dashboard Tabs

4

User Role Types

10+

Booking Steps

24/7

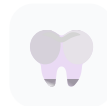
Bot Availability

Key Value Propositions



WhatsApp-Native

Patients book through WhatsApp with no app download. Works on any phone, even 2G.



Multi-Clinic Ready

Each clinic gets its own isolated environment. One platform serves unlimited dental clinics.



Intelligent Bot



Auto Reminders

Conversational AI guides patients step-by-step.
Handles booking, reschedule, and cancel.

Sends WhatsApp reminders 24 hours and 2 hours
before each appointment automatically.



Rich Analytics

30-day trends, heatmaps, no-show rates, and
doctor performance — all in real-time.



Secure by Design

Bank-grade encryption, audit logs, rate limiting,
and isolated databases per clinic.



How it works in one sentence: A patient sends "Hi" to the clinic's WhatsApp number → the MediBook bot guides them through selecting a doctor, date, and time → the appointment is confirmed automatically, no human intervention needed.

2. User Roles & Personas

MediBook has four distinct user types, each with purpose-built interfaces and carefully scoped permissions.



Super Admin

The platform operator who manages the entire MediBook SaaS business.

- ✓ Create and manage clinic tenants
- ✓ Assign and change subscription plans
- ✓ Suspend or reactivate any clinic
- ✓ View platform-wide revenue & stats
- ✓ Monitor health of all clinic instances
- ✓ Manage pricing plans and features



Clinic Admin

The owner or manager of an individual dental clinic.

- ✓ Manage dentists, schedules, and leaves
- ✓ View and manage all appointments
- ✓ Access analytics and calendar view
- ✓ Configure clinic settings & branding
- ✓ Add and manage clinic staff
- ✓ View patient records and history
- ✓ Test the WhatsApp bot directly
- ✓ View complete audit logs



Clinic Staff

Receptionists and nurses with day-to-day appointment access.

- ✓ View and update appointments
- ✓ Look up patient records
- ✓ Mark appointments as completed
- ✓ Add notes to appointments



Patient

The end customer who interacts exclusively via WhatsApp. No login required.

- ✓ Book appointments via WhatsApp
- ✓ View their upcoming appointments
- ✓ Reschedule or cancel bookings
- ✓ Receive automated reminders

- ✓ Cannot change clinic settings
- ✓ Cannot access financial data

- ✓ Leave feedback and ratings
- ✓ Opt-out of messages anytime

Access Matrix

FEATURE / CAPABILITY	SUPER ADMIN	CLINIC ADMIN	STAFF	PATIENT
Book Appointment	—	✓ Manual	✓ Manual	✓ WhatsApp
View All Appointments	X	✓	✓	✓ Own only
Manage Doctors	X	✓	X	X
View Analytics	✓ Platform	✓ Clinic	X	X
Manage Tenants/Clinics	✓	X	X	X
Change Settings	✓ Platform	✓ Clinic	X	X
View Audit Logs	✓	✓	X	X
Manage Staff	X	✓	X	X

3. Patient Journey — WhatsApp Bot

The patient never needs to install anything. The entire experience happens through a conversation on WhatsApp.

3.1 Booking an Appointment

From the first message to a confirmed booking takes approximately 2–3 minutes.

BOOKING FLOW — STEP BY STEP

1

Patient sends "Hi"

Patient messages the clinic's WhatsApp number. The bot instantly responds with a welcome greeting and a main menu with options: Book Appointment, My Appointments, Help.

2

Select Hospital (if multiple branches)

If the clinic has multiple locations, the patient selects which hospital they want to visit. Single-branch clinics skip this step automatically.

3

Select Treatment / Specialty

A list of dental treatment categories is shown (e.g., General Dentistry, Root Canal Treatment, Orthodontics & Braces). Patient picks the relevant specialty. All appointments are always in-clinic.

4

Choose Dentist

Available dentists in the chosen specialty are listed with their qualifications and consultation fees. Patient selects their preferred dentist.

5

Select Date

The next 7 available days (with open slots) are presented as options. Days without availability are automatically excluded.

6

Choose Time Slot

Available time slots for the selected date are shown (e.g., 10:00 AM, 10:30 AM). Fully booked slots are not displayed.

7

Reason for Visit (Chief Complaint)

Patient selects their primary reason for the visit: Pain/Emergency, Checkup/Cleaning, or Cosmetic/Other. This is saved with the appointment for the dentist's reference.

8

Enter Patient Details (First-Time Only)

New patients provide their full name, date of birth, and gender. Returning patients are recognized by phone number and skip this step.

9

Confirm Booking

A full summary is shown: Dentist name, clinic, date, time, reason for visit. Patient confirms with a single tap.

✓

Appointment Confirmed! 🎉

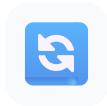
A unique **Booking ID** (e.g., MB12AB3CD) is sent to the patient. This ID is used for rescheduling, cancellation, and reference. The slot is locked instantly.

3.2 Managing Existing Appointments



View Appointments

Patient types "My Appointments" to see a list of upcoming and past bookings with dates and doctor names.



Reschedule

Patient selects a booking, picks a new date and slot. Old slot is released, new slot is reserved atomically.



Cancel

Patient selects a booking and confirms cancellation with a reason. The slot becomes available for others immediately.

3.3 Reminders & Follow-Ups



24-Hour Reminder

Automatic WhatsApp message sent the day before the appointment with all appointment details.



2-Hour Reminder

Final reminder sent 2 hours before the appointment. Reduces no-shows significantly.



Post-Visit Feedback

After the appointment date, the bot asks for a rating (1–5 stars) and optional comment.



Waitlist Notification

If fully booked, patients can join a waitlist and receive a WhatsApp notification when a slot opens up.

3.4 Opt-Out & Privacy



Patients can type **STOP** or **UNSUBSCRIBE** at any time to stop receiving all messages from the clinic. They can re-enable messages by typing **START**. This is fully automated and compliant with WhatsApp's messaging policies.

4. Clinic Admin Dashboard

The admin dashboard is a powerful, browser-based control center with 14 dedicated tabs — each serving a specific management need.

4.1 Dashboard Overview

Upon login, the admin sees a live overview of the clinic's operations with four key metrics prominently displayed:



Below the stats, **Today's Schedule** shows the next 10 upcoming appointments with patient name, doctor, and time — giving the front desk a quick morning brief at a glance.

4.2 All 14 Dashboard Tabs

1 Overview Live stat cards, today's schedule, and quick action shortcuts.	2 Appointments Full appointment list with filters by date, status, doctor. Includes CSV export.
3 Doctors Add/edit/deactivate doctors. Set qualifications, fees, slot durations.	4 Clinics Manage clinic branches. Add address, phone, city for each location.
5 Patients Searchable patient directory. View full history, visit count, medical notes.	6 Waiting List Patients waiting for specific doctors. Send them a WhatsApp notification with one click.

7

Feedback

Patient reviews and star ratings after visits. Filter by doctor or date range.

8

Analytics

30-day trends, heatmaps, top doctors, and appointment status breakdown.

9

Calendar

Monthly calendar view showing appointment volume per day. Click any day for details.

10

Slots

Generate time slots, block date ranges for holidays, view slot availability.

11

Staff

Add/remove clinic staff accounts. Manage roles and permissions.

12

Leaves

Record doctor leave days. Automatically blocks affected time slots.

13

Settings

Clinic name, address, notifications, WhatsApp credentials, language preferences.

14

Bot Tester

Simulate patient WhatsApp conversations directly in the browser for testing.

4.3 Doctor Management

Each doctor profile stores:

**Profile Information**

Name, specialization, qualifications, dental treatment department, and affiliated clinic branch.

**Weekly Schedule**

Working hours for each day of the week, including lunch break windows and days off.

**Fees & Slot Duration**

Consultation fee (shown to patients on WhatsApp) and appointment slot length (e.g., 15, 30, or 45 minutes).

4.4 Appointment Management

Every appointment in the system has:

FIELD	DESCRIPTION
Booking ID	Unique alphanumeric ID (e.g., MB12AB3CD) shared with the patient
Status	Confirmed → Completed / Cancelled / No-Show
Chief Complaint	Reason for visit (Pain/Emergency, Checkup/Cleaning, Cosmetic/Other)
Notes	Staff can add clinical notes with category tags
Cancellation Info	Who cancelled (patient/admin/bot) and reason
Reminder Status	Whether 24h and 2h reminders were sent

4.5 Analytics Dashboard

The Analytics tab provides visual reports to help clinics understand performance:



30-Day Trend

Line chart showing daily appointment volume for the past month. Identifies busy and slow periods.



Top Doctors

Bar chart ranking doctors by appointment volume. Helps with resource planning.



Status Breakdown

Pie/donut split of Confirmed, Completed, Cancelled, and No-Show appointments.



Heatmap

7-day appointment distribution by hour. Reveals peak booking times.



Summary KPIs

Total confirmed appointments, no-show rate percentage, average consultation fee.



CSV Export

Export filtered appointment data to spreadsheet for offline reporting.

4.6 Slot Generation

Time slots are the building blocks of the booking system. Each slot represents one bookable appointment window.



Automatic Nightly Generation: Every night, the system automatically generates slots for the next 60 days for all active doctors based on their weekly schedule. No manual action required from admins.



Manual Override: Admins can manually generate slots for a specific doctor, block a date range (e.g., public holiday, doctor vacation), or toggle individual slots between available and blocked.

4.7 Audit Logs

Every action taken by any admin or staff member is permanently recorded:

WHAT IS LOGGED	INFORMATION CAPTURED
Login / Logout events	User email, IP address, browser, timestamp
Appointment changes	Who changed it, what changed (before/after), when
Doctor profile edits	Previous and new values for every field
Patient record access	Who viewed or modified patient data
Settings changes	Configuration changes with old and new values
Staff management	Account creation, deactivation, password changes

4.8 Onboarding Checklist

New clinics are guided through a step-by-step setup checklist on first login:

1

Add Clinic Branch

Register at least one clinic location with address and contact details.

2

Create Departments

Add dental treatment categories (e.g., General Dentistry, Root Canal Treatment, Orthodontics).

3

Add Doctors

Create doctor profiles with their weekly schedule and consultation fees.

4

Connect WhatsApp

Enter the Meta WhatsApp Business API credentials to activate the bot.

✓

Go Live!

The bot is active. Patients can now book appointments via WhatsApp.

5. Platform Super Admin

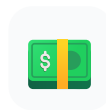
The Super Admin manages the entire MediBook SaaS business from a dedicated control panel separate from individual clinics.

5.1 Platform Dashboard



Tenant Overview

Total clinics on the platform, active vs suspended count, new signups this month.



Revenue (MRR)

Monthly Recurring Revenue calculated from active subscriptions across all plans.



Plan Distribution

Breakdown of how many clinics are on each pricing plan (Starter / Growth / Professional / Enterprise).

5.2 Tenant Management

Creating a new clinic tenant automatically:

①

Creates a Dedicated Database

A completely isolated database schema is provisioned for the new clinic.

②

Creates Admin Account

A clinic admin account is created with the provided email and password.

③

Assigns Clinic ID (Slug)

A unique URL-friendly identifier (e.g., "city-heart-clinic") is assigned for login.

④

Clinic is Ready

The clinic admin can immediately log in and begin the onboarding checklist.

5.3 Clinic Health Monitoring

For every clinic, the Super Admin can run a real-time health check:

Database Status

- Connection responsive
- Schema exists and is valid
- No migration errors

Message Queue

- Queue depth (pending messages)
- Worker status
- Last processed timestamp

Cron Health

- Last slot generation run
- Last reminder batch
- Alert if any cron missed

Usage Stats

- Appointments this month
- Total patients registered
- Last activity timestamp

6. Automation & Background Tasks

MediBook runs several automated jobs in the background 24/7 — invisible to users but essential to the platform's reliability.

 11:30 PM IST · Daily

Slot Generator

Generates appointment time slots for all active doctors for the next 60 days, based on their configured weekly schedule and lunch breaks. Runs even if admin forgets to do it manually.

 Every Hour

24h & 2h Reminders

Scans all confirmed appointments. Sends WhatsApp reminders to patients with appointments in ~24 hours and ~2 hours. Prevents no-shows without any admin action.

 Every 6 Hours

Backup Reminder Runner

Catches any reminders that were missed due to network failures or WhatsApp API rate limits. Ensures delivery reliability for all patients.

 Real-Time

WhatsApp Message Queue

Incoming patient messages are queued asynchronously and processed in order. The queue handles traffic spikes gracefully and retries failed messages automatically.



Duplicate Prevention: When multiple servers run simultaneously, a distributed locking system ensures each cron job runs exactly once — no duplicate reminders sent, no double-generated slots.

7. Security & Data Protection

MediBook is built with a security-first mindset. Patient data is protected at every layer.

Authentication

- Passwords hashed with bcrypt (12 rounds)
- JWT tokens expire in 24 hours
- 30-day refresh tokens (one-time use)
- Token blacklist on every logout
- Password reset via secure email link (1-hour expiry)

Data Isolation

- Each clinic has its own database schema
- Zero chance of cross-clinic data leakage
- Tenant ID validated on every API request
- WhatsApp credentials encrypted (AES-256)

Rate Limiting

- Login: 20 attempts per 15 minutes
- Forgot password: 5 requests per hour
- Slot generation: 5 per hour
- Analytics: 20 requests per minute
- Webhook: 2,000 messages per minute

Web Security Headers

- HTTPS enforced (HSTS 1 year)
- Content Security Policy (CSP)
- Clickjacking protection (X-Frame-Options)
- MIME type sniffing prevention

Input Validation

- All API inputs validated before processing
- Phone numbers: 7–20 digit format enforced
- UUIDs validated on all ID parameters
- Message length limits on bot inputs

Audit Trail

- Every admin action permanently logged
- Before/after values stored for all edits
- IP address and browser recorded
- Cannot be modified by any user



Patient Data Privacy: Patient WhatsApp conversations and medical history are stored only within the clinic's isolated database. No data is shared between clinics, and patients can opt-out of all messages at any time by texting **STOP**.

8. Subscription Plans

MediBook offers four tiered plans to suit dental clinics of every size — from solo dentists to multi-branch dental chains.

STARTER

₹999/mo

BEST FOR SOLO

- ✓ Up to 3 doctors
- ✓ 100 appointments/month
- ✓ WhatsApp booking bot
- ✓ Basic analytics
- ✓ Email support

GROWTH

₹2,499/mo

MOST POPULAR

- ✓ Up to 10 doctors
- ✓ 500 appointments/month
- ✓ All Starter features
- ✓ Advanced analytics
- ✓ Waiting list management
- ✓ Priority support

PROFESSIONAL

₹4,999/mo

FOR CLINICS

- ✓ Up to 25 doctors
- ✓ 2,000 appointments/month
- ✓ All Growth features
- ✓ Staff accounts
- ✓ Audit logs
- ✓ Dedicated support

ENTERPRISE

Custom

DENTAL CHAINS

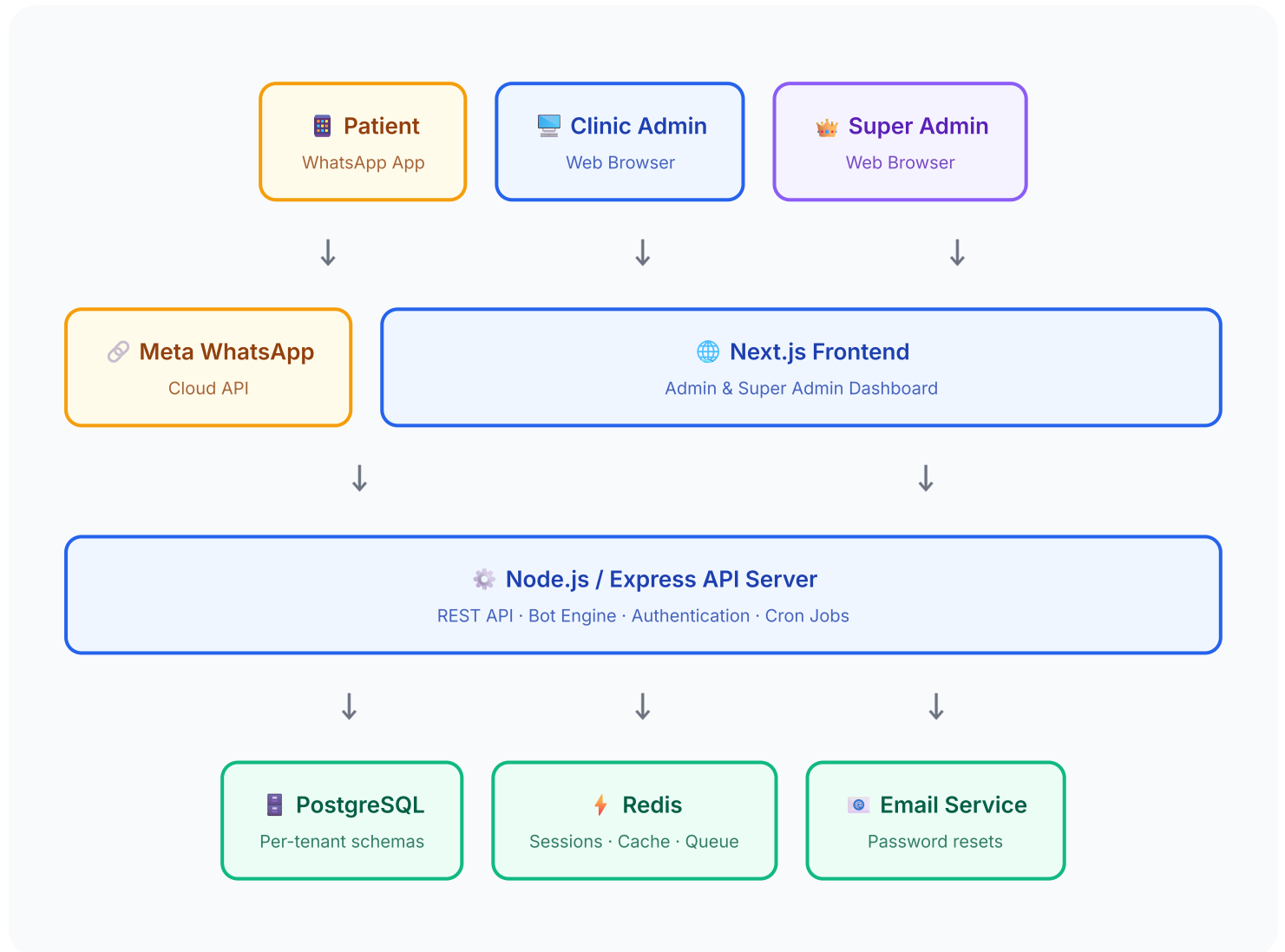
- ✓ Unlimited dentists
- ✓ Unlimited appointments
- ✓ All Professional features
- ✓ Custom integrations
- ✓ SLA guarantee
- ✓ Dedicated account manager



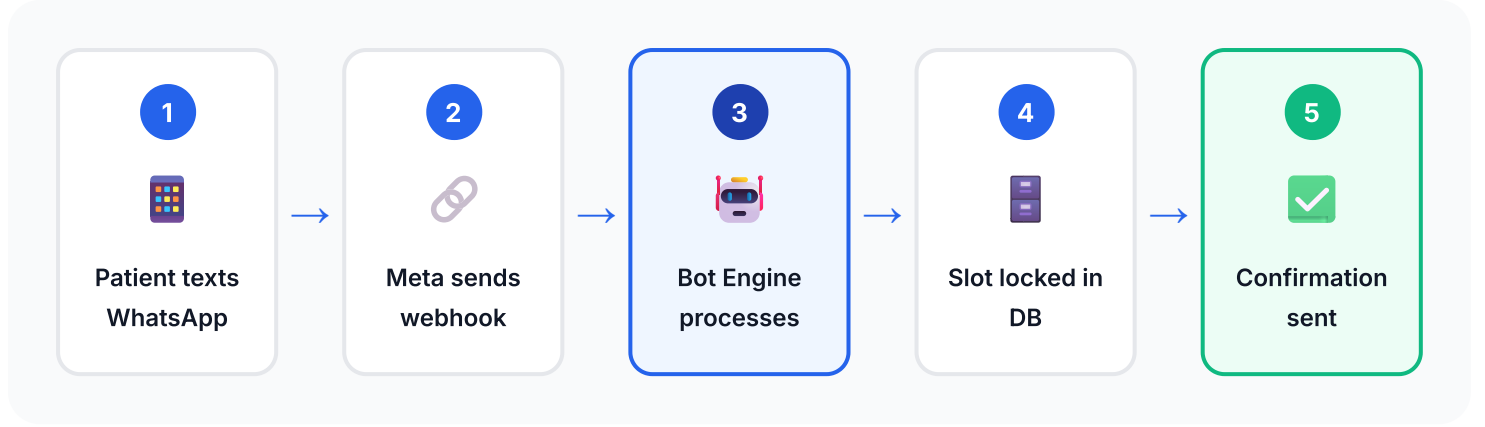
Plan upgrades and downgrades are managed by the Super Admin. Changes take effect immediately. All existing data is preserved when changing plans.

9. System Architecture

A simplified view of how MediBook's components connect and communicate.



Data Flow: Patient Booking



10. WhatsApp Conversation Flows & Message Templates

This section documents every message the bot sends — the exact text, message type, trigger condition, and conversation state — extracted directly from the source code.

10.1 WhatsApp Message Types Used

MediBook uses three native WhatsApp message types. Each is chosen based on the number of options and WhatsApp platform limits:



Text Message

Plain text with ***bold*** and *_italic_* markdown. Used for confirmations, errors, reminders, and any message with more than 3 options (numbered list fallback).

API type: text



Interactive Buttons

Up to **3 tappable reply buttons**. Used for yes/no decisions, main menu, gender selection, and visit type. Button titles are max 20 characters.

API type: interactive / button



Interactive List

A scrollable dropdown with sections and rows. Used for selecting hospitals, specialties, doctors, dates, and time slots. Falls back to numbered text if WhatsApp rejects it.

API type: interactive / list



Fallback behaviour: If an interactive message fails to send (e.g. unsupported WhatsApp version), the bot automatically falls back to a plain numbered text list — so the patient can always reply with "1", "2", "3" etc.

10.2 Trigger Keywords & Shortcuts

These keywords work at any point in a conversation regardless of the current state:

KEYWORD(S)	ACTION	WORKS WHEN
Hi, Hello, Hey, Menu, Start, Hy, Hai	Reset to Main Menu with welcome message	Any state
STOP, UNSUBSCRIBE, OPT-OUT, BLOCK	Opt out of all messages from the clinic	Any state
START	Re-subscribe to receive messages again	After opting out
RESCHEDULE	Jump directly to reschedule flow (skip menu)	Any state
CANCEL APPOINTMENT	Jump directly to cancel flow (skip menu)	Any state
BOOK	Start booking flow from main menu	MAIN_MENU state
STATUS	Check appointment status by Booking ID	MAIN_MENU state
SKIP	Skip optional fields (email, feedback comment)	COLLECT_EMAIL, feedback states
Help, Info	Show help menu with available commands	MAIN_MENU state

10.3 Complete Bot State Machine

The bot tracks exactly where each patient is in their conversation using a state stored per phone number. There are **20 distinct states** (visit type selection removed; chief complaint step added for dental):

STATE	DESCRIPTION	FLOW
idle	No active conversation. Waiting for "Hi".	All flows
main_menu	Main menu shown — waiting for choice.	Entry
select_hospital	Waiting for patient to pick a clinic branch.	Booking
select_department	Waiting for dental treatment/specialty selection.	Booking
select_doctor	Waiting for dentist selection.	Booking
select_date	Waiting for appointment date.	Booking
select_slot	Waiting for time slot selection.	Booking
collect_name	New patient — asking for full name.	Booking
collect_dob	New patient — asking for date of birth (DD/MM/YYYY).	Booking
collect_gender	New patient — asking for gender.	Booking
collect_email	Asking for email (optional, for confirmation email).	Booking
collect_chief_complaint	Asking reason for visit: Pain/Emergency, Checkup, or Cosmetic.	Booking
confirm_booking	Summary shown — waiting for Confirm or Cancel.	Booking
check_booking_status	Waiting for Booking ID to look up status.	Status Check
my_appointments	Appointment list shown — waiting for action.	Management
reschedule_select	Waiting for Booking ID to reschedule.	Reschedule
reschedule_date	Waiting for new date selection.	Reschedule
reschedule_slot	Waiting for new time slot.	Reschedule
reschedule_confirm	Summary shown — waiting for confirm/keep.	Reschedule
cancel_select	Waiting for Booking ID to cancel.	Cancel

STATE	DESCRIPTION	FLOW
cancel_reason	Waiting for cancellation reason.	Cancel
cancel_confirm	Waiting for final cancel confirmation.	Cancel
collect_feedback_rating	Post-visit — waiting for 1–5 star rating.	Feedback
collect_feedback_comment	Rating received — waiting for comment or Skip.	Feedback

10.4 Booking Flow — Exact Message Templates

Every message below is extracted verbatim from the source code. Emoji, bold (*text*), and line breaks are exactly as the patient sees them on WhatsApp.

① Welcome / Main Menu

idle → main_menu

Trigger: Patient sends "Hi", "Hello", "Hey", "Menu", "Start" — or any message when state is idle.

💬 BOT → PATIENT (INTERACTIVE BUTTONS)

Interactive Buttons

👋 Welcome, **Rahul** to **Demo Clinic Hyderabad!**

How can I help you today?



Book Appointment



My Appointments



Check Status



The greeting is personalised: returning patients see their first name. New patients see a generic subtitle: *"I can help you book appointments, check your status, or manage existing bookings."*

② Hospital Selection (only shown if clinic has multiple branches)

main_menu → select_hospital

Trigger: Patient taps "📅 Book Appointment" and clinic has more than 1 active hospital.

💬 BOT → PATIENT (INTERACTIVE LIST)

Interactive List

Select a Clinic

Choose your preferred location:

 [View Locations](#)

List opens with section "Our Locations":

- Smile Dental Clinic — Banjara Hills
- Smile Dental Clinic — Secunderabad
- (all active clinic branches)

③ Treatment / Specialty Selection

`select_hospital` → `select_department`

Trigger: Hospital selected (or auto-selected if only one branch). All dental appointments are always in-clinic — no visit type step required.

 BOT → PATIENT (BUTTONS IF ≤3 / LIST IF 4+)

Interactive Buttons (≤3 departments)

Select Treatment

What brings you in today?

General Dentistry

Root Canal Treatment

Orthodontics & Braces

Interactive List (4+ departments)

Select Treatment

What brings you in today?

 [View Treatments](#)

④ Dentist Selection

`select_department` → `select_doctor`

Trigger: Treatment selected. Dentist list includes qualification and consultation fee.

 BOT → PATIENT (LIST WITH DENTIST DETAILS)

Interactive List

 **Select Dentist**

 [View Dentists](#)

Section "Root Canal Treatment Dentists":

Dr. Priya Sharma

BDS, MDS (Endodontics) • ₹800

Dr. Anita Reddy

BDS, MDS (Oral Surgery) • ₹600



No slots available? Bot shows a message to try a different date or dentist.

⑤ Date Selection

`select_doctor` → `select_date`

Trigger: Dentist selected. Shows next 7 dates that have available slots (today excluded if no future slots today).

 BOT → PATIENT (INTERACTIVE LIST)

Interactive List

 **Select Date**

Available dates for Dr. Priya Sharma:

 **Choose Date**

Section "Available Dates":

Mon, 17 Mar — 16 slots available

Tue, 18 Mar — 16 slots available

Wed, 19 Mar — 14 slots available

...up to 7 dates shown

10.4 Booking Flow (continued)

⑦ Time Slot Selection

select_date → select_slot

Trigger: Date selected. If selected date has no slots, suggests next 3 available dates instead of failing silently.

💬 BOT → PATIENT (INTERACTIVE LIST)

Interactive List

 **Select Time**

Slots on Mon, 17 Mar:

 **Choose Time**

Section "Available Slots":

09:00 — 09:00 – 09:30

09:30 — 09:30 – 10:00

10:00 — 10:00 – 10:30

...all available slots for that day



No slots on selected date: "No slots available on that date. 📅 Next available dates for Dr. Priya Sharma: • Wed, 19 Mar (14 slots) • Thu, 20 Mar (16 slots). Reply Hi to go back and choose a date."

⑧–⑪ Patient Details (New Patients Only)

select_slot → collect_name → collect_dob → collect_gender → collect_email

Trigger: Slot selected and patient is not in the database. Returning patients skip directly to 12.

COMPLETE PATIENT DETAIL COLLECTION FLOW

Your Name

Please enter your full name:

Rahul Sharma

Date of Birth

Enter your DOB in DD/MM/YYYY format:

Example: 15/08/1990

15/08/1990

Interactive Buttons

Your Gender

Male

Female

Other

Male

Email Address *(optional)*

Share your email to receive a booking confirmation:

Or reply **Skip** to continue without it.

rahul@gmail.com



Validation errors: Name must be 2–100 characters. DOB must be DD/MM/YYYY (1900–today). Invalid input repeats the prompt: *"Invalid format. Please use DD/MM/YYYY. Example: 15/08/1990"*

collect_email → confirm_booking

Trigger: All patient details collected (or returning patient identified). Shows full summary before committing.

💬 BOT → PATIENT (INTERACTIVE BUTTONS)

Interactive Buttons

📋 **Please review your booking**

🏥 **Clinic:** Smile Dental Clinic

🦷 **Treatment:** Root Canal Treatment

👩 **Dentist:** Dr. Priya Sharma

📅 **Date:** Monday, 17 March 2025

🕒 **Time:** 09:00

📄 **Reason:** 🚨 Pain / Emergency

💰 **Consultation Fee:** ₹800

👤 **Patient:** Rahul Sharma

Tap **Confirm** to book your appointment.

✅ Confirm

❌ Cancel

⑬ **Booking Confirmed** ✅

confirm_booking → idle

Trigger: Patient taps "✅ Confirm". Slot is atomically locked in the database. A unique Booking ID is generated.

💬 BOT → PATIENT (TEXT — FINAL CONFIRMATION)

🎉 **Appointment Confirmed!**

Your booking is all set. Here are your details:

📄 **Booking ID:** MB12AB3CD

👩 **Dr. Priya Sharma**

🏥 **Smile Dental Clinic**

📅 **Mon, 17 Mar 2025**

🕒 **09:00**

📌 **Please save your Booking ID.** You'll need it to reschedule or cancel.

👉 **Before you head in:** Avoid eating 1 hour before · Bring X-rays/records ·
Declare medications or allergies · Arrive 10 min early

We'll remind you 24 hours before. See you then! 😊



Simultaneous actions on confirmation: (1) Patient gets this WhatsApp message. (2) Patient email gets a booking confirmation email (if email was provided). (3) Clinic admin gets a new booking alert email. (4) The slot is locked atomically — no double-booking possible.



Race condition — slot taken: If another patient books the same slot in the split second between selection and confirmation, the patient sees: "⚠️ *Slot no longer available. Someone just booked that slot. Please choose a different time. Reply Hi to go back to the menu.*"

10.5 Appointment Management — Exact Message Templates

① My Appointments List

main_menu → my_appointments

Trigger: Patient taps "📅 My Appointments". Shows up to 5 upcoming and 3 past appointments.

 BOT → PATIENT (INTERACTIVE BUTTONS)

Interactive Buttons

 **Upcoming Appointments**

1. **MB12AB3CD**

 Dr. Priya Sharma

 Mon, 17 Mar at 09:00

 Confirmed

 **Past Appointments**

• **MB09XY2ZA** — Dr. Rajesh Kumar, 10 Feb 2025 —  Completed

What would you like to do?

 Reschedule

 Cancel Appointment

 Main Menu

 **No appointments found:** "We don't have any appointments linked to this number. Reply Hi and tap Book Appointment to schedule your first one! 😊"

② Reschedule Flow (4 steps)

my_appointments → reschedule_select → reschedule_date → reschedule_slot → reschedule_confirm → idle

 COMPLETE RESCHEDULE CONVERSATION

Enter the **Booking ID** to reschedule (e.g. MB12AB3):

MB12AB3CD

Interactive List

 Reschedule — MB12AB3CD

Currently: Dr. Priya Sharma on 17 Mar at 09:00

Pick a **new date**:

 **Choose New Date**

[Selects Tue, 18 Mar]

Interactive List

 Select New Time

Slots on Tue, 18 Mar:

 **Choose Time**

[Selects 10:30]

Interactive Buttons

 Confirm Reschedule

Dr. Priya Sharma

 Old: Mon, 17 Mar at 09:00

 New: Tue, 18 Mar at 10:30

Confirm the change?

✓ Yes, Reschedule

✗ No, Keep Original

Yes, Reschedule

✓ **Appointment Rescheduled!**

Your booking has been moved successfully.

👤 Booking ID: **MB12AB3CD**

👩 Dr. Priya Sharma

📅 Tue, 18 Mar 2025

🕒 10:30

We'll send you a fresh reminder 24 hours before. See you then! 😊



2-hour policy: If the appointment is less than 2 hours away, reschedule is blocked: "⚠️
Rescheduling must be done at least 2 hours before the appointment. Your appointment is in less than 2 hours. Please call the clinic directly."

③ Cancellation Flow (3 steps)

my_appointments → cancel_select → cancel_reason → cancel_confirm → idle

💬 COMPLETE CANCELLATION CONVERSATION

Enter the **Booking ID** to cancel (e.g. MB12AB3):

MB12AB3CD

Interactive Buttons — Reason Collection

✗ **Cancel Appointment**

Booking: **MB12AB3CD**

👩 Dr. Priya Sharma

📅 Mon, 17 Mar at 09:00

Before we cancel — could you tell us why?

Doctor not available

Schedule conflict

Other reason

Schedule conflict

Interactive Buttons — Final Confirmation

✖ Confirm Cancellation

Booking: **MB12AB3CD**

👤 Dr. Priya Sharma

📅 Mon, 17 Mar at 09:00

📄 Reason: Schedule conflict

This cannot be undone. Are you sure?

Yes, Cancel It

No, Keep It

Yes, Cancel It

✅ Appointment Cancelled

Your appointment has been cancelled and the slot released.

We hope everything is okay. Whenever you're ready, reply **Hi** to book again.



After cancellation, the slot is immediately released and made available for other patients. If any patients are on the waitlist for that doctor, they automatically receive a "Slot Available!" notification.

10.6 Automated Message Templates (System-Sent)

These messages are sent automatically by the system — not triggered by patient input. They use WhatsApp Approved Message Templates via the Meta Cloud API.



WhatsApp Template Messages: Unlike conversational messages, these are pre-approved templates required by Meta for business-initiated messages outside a 24-hour window. The bot uses `sendTemplate()` for reminders, with a plain-text fallback if the template hasn't been approved yet.

① 24-Hour Appointment Reminder

Sent: Automatically every hour, to all patients with appointments **tomorrow** whose 24h reminder has not yet been sent.

Template name: `appointment_reminder_24h` | **Cron:** Hourly at :00

💬 BOT → PATIENT (WHATSAPP APPROVED TEMPLATE)

Appointment Reminder

You have a dental appointment tomorrow!

 Dr. Priya Sharma

 Tue, 18 Mar at 10:30

 Smile Dental Clinic

Reply **Hi** to reschedule or cancel.

Template variables: `{{1}}` = dentist name, `{{2}}` = date, `{{3}}` = time, `{{4}}` = clinic name

② 2-Hour Appointment Reminder

Sent: Every hour, to all patients with appointments today within the next 2 hours whose 2h reminder has not been sent.

Template name: `appointment_reminder_2h` | Fallback to plain text if template not approved.

💬 BOT → PATIENT (TEMPLATE WITH PLAIN-TEXT FALLBACK)

 **Heads up — appointment in 2 hours!**

 Dr. Priya Sharma

 10:30

 **Quick checklist before you go:**

- Arrive 10 minutes early
- Bring any previous reports or prescriptions
- Carry a valid photo ID
- Have a list of your symptoms or questions ready

Need to make changes? Reply **Reschedule** or **Cancel Appointment**.

Template variables: {{1}} = doctor name, {{2}} = appointment time

③ Waitlist Slot Available Notification

Sent: Automatically when a booking is cancelled or a slot becomes available, to the first 5 patients on the waitlist for that doctor.

 BOT → WAITLISTED PATIENT (TEXT)

 **Slot Available!**

A slot has opened up with Dr. Priya Sharma.

Reply **Hi** to book your appointment now before it's taken!



When the notification is sent, the patient's bot session is automatically moved to `main_menu` state — so as soon as they reply, they immediately see the booking options.

④ Post-Visit Feedback Request

Sent: Daily at 10:00 AM IST, to patients whose appointment was completed or marked no-show the previous day, and who have not yet submitted feedback.

 BOT → PATIENT (TEXT — TRIGGERS FEEDBACK FLOW)

 **How did it go, Rahul?**

We hope your visit with Dr. Priya Sharma went well!

Rate your experience:

1 ★ — Poor

2 ★★ — Below average

3 ★★★ — Average

4 ★★★★ — Good

5 ★★★★★ — Excellent

Just reply with a number **1–5**. Takes 5 seconds! 🙏

10.7 Feedback Collection Flow

Feedback — Rating + Comment

collect_feedback_rating → collect_feedback_comment → idle

Trigger: Patient replies with a number 1–5 after receiving the feedback request.

PATIENT RATES 5 STARS

5

🌟 5/5 — Amazing!

Would you like to add a comment? Your feedback helps us improve.

(Reply Skip to finish)

Very professional and caring doctor!

✅ Thank you for your feedback!

It genuinely helps the clinic improve. We appreciate you taking the time. 🙏

Reply **Hi** for the main menu.

RATING	RESPONSE TEXT	EMOJI
★★★★★ 5	"Amazing!"	🌟
★★★★☆ 4	"Great to hear!"	🌟
★★★☆☆ 3	"Thanks for the feedback."	📄✍️
★★☆☆☆ 2	"Sorry to hear that."	📄✍️

RATING	RESPONSE TEXT	EMOJI
★ 1	"Sorry to hear that."	📄

10.8 Other System Messages

 Opt-Out Confirmation <i>"You have been unsubscribed from [Clinic Name] WhatsApp notifications. To re-subscribe, reply START."</i>	 Booking Status Check Patient sends "Status" → enters Booking ID → sees status, doctor, date, time with emoji: ✅ Confirmed / ❌ Completed / ❌ Cancelled / ⚠️ No Show.
 Unrecognised Input Fallback <i>"I didn't quite get that. 🤖 Here's what you can do: Reply Hi — Main menu / Reply Book — Book an appointment / Reply Status — Check appointment status / Reply Cancel Appointment — Cancel a booking"</i>	 Help Menu <i>"📘 Help • Reply Book — Book an appointment • Reply Status — Check appointment status • Reply My — View your appointments • Reply Hi — Return to main menu. For emergencies, please call the clinic directly."</i>
 Rate Limit Warning <i>"⚠️ You've made 3 bookings in the last hour. Please wait before booking again. Reply Hi for the main menu." (Max 3 bookings per hour per phone number)</i>	 Session Error Recovery <i>"Sorry, something went wrong. Let's start over — reply Hi to continue." Session is reset to idle. Prevents infinite error loops.</i>

10.9 Message Audit Logging

Every inbound and outbound WhatsApp message is logged to the `wa_messages` table for audit and compliance purposes:

FIELD	DESCRIPTION	EXAMPLE
phone	Patient's WhatsApp number	919876543210
direction	in (patient→bot) or out (bot→patient)	out

FIELD	DESCRIPTION	EXAMPLE
message_type	text / buttons / list	buttons
content	Full message text	 Appointment Confirmed!...
wa_message_id	Meta's unique message ID for delivery tracking	wamid.HBg...
created_at	Timestamp	2025-03-17 09:01:23 IST



Duplicate prevention: Inbound messages are deduplicated by `wa_message_id`. If Meta delivers the same webhook twice (which can happen), the second delivery is silently ignored, preventing double-processing of bookings.

11. Getting Started

Everything needed to run MediBook is included. The platform can be up and running in under 30 minutes.

10.1 Demo Credentials

ROLE	LOGIN URL	EMAIL	PASSWORD	CLINIC ID
SUPER ADMIN	/login → Super Admin tab	admin@medibook.com	SuperAdmin@123	—
CLINIC ADMIN	/login → Clinic Admin tab	demo@medibook.com	Demo@123456	demo-clinic

10.2 Test the WhatsApp Bot (No WhatsApp Needed)

The **Bot Tester** tab in the admin dashboard lets you simulate patient conversations directly in the browser. Enter a test phone number, type a message (e.g., "Hi"), and see exactly what patients experience — without needing real WhatsApp credentials.



Instant Bot Testing: Navigate to Dashboard → Bot Tester tab. Enter phone number 919999999999 and message Hi. Click Send to see the full booking flow.

10.3 Setting Up a New Clinic

1

Super Admin creates tenant

Logs into Super Admin panel, creates new tenant with clinic name, slug, and owner credentials.

2

Clinic Admin logs in

Using provided credentials and clinic slug at the login page.

3

Complete onboarding checklist

4

Share WhatsApp number with patients

The clinic's WhatsApp number is live. Patients can now book appointments instantly.

10.4 Technical Requirements

COMPONENT	TECHNOLOGY	PURPOSE
Backend API	Node.js + Express	Business logic, bot engine, cron jobs
Frontend	Next.js (React)	Admin dashboard and super admin panel
Database	PostgreSQL	All clinic and patient data (schema-per-tenant)
Cache / Queue	Redis + BullMQ	Session cache, message queue, cron locking
WhatsApp	Meta Cloud API	Send and receive WhatsApp messages
Email	Resend	Password reset emails
Deployment	Docker Compose	One-command local and production deployment



WhatsApp Appointment Booking Platform for Indian Healthcare

Developed & Powered by Swalambha AI Technologies

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